

Data Cleaning Report

I. Books Dataset

i. Remove duplicates

- Checked for duplicate records using `duplicated().sum()`.
- No duplicated records as found.
- This ensured that only unique book records are present.

ii. Handle missing values:

- Missing values in Book-Author, Publisher, and image URLs were filled with 'Unknown'.
- This helped retain records without dropping useful information.

iii. Standardize publication years:

- Converted Year-Of-Publication to numeric format.
- Replaced invalid years (like <1800 or >2025) with NaN.
- Filled missing values with the column median.
- Converted cleaned values to int.

iv. Summary:

Metric	Before	After
Rows	271360	271360
Missing values	In Publisher, Book-Author, Image-URL-L	Filled
Duplicate entries	0	0

v. Decisions:

- publisher , author, Image url were filled with 'Unknown' to retain records.
- Publication years were cleaned to remove outliers and inconsistencies.

II. Users Dataset

i. Remove duplicate records

- Checked for duplicate records using `duplicated().sum()`.
- No duplicated records as found.
- This ensured that only unique user records are present.

ii. Handle missing values:

- Retrieved only rows where age is either missing or between 5 and 100.
- Filled missing Age values with the median age.
- This prevented unrealistic age values from affecting analysis.

iii. Summary:

Metric	Before	After
Rows	278858	278858
Missing values	In Age	Filled
Duplicate entries	0	0

iv. Decisions:

- Unrealistic age entries were removed for quality.
- Missing age values were filled with median to retain records without adding bias.

III. Ratings Dataset

i. Remove duplicate

- Checked for duplicate records using duplicated().sum().
- No duplicated records as found.
- This ensured that only unique Rating records are present.

ii. Handled missing values:

- No missing values are found.

iii. Summary:

Metric	Before	After
Rows	1149780	1149780
Missing Values	No	No
Duplicate record	0	0